

Cooperation in Virtual Reality: Exploring Environmental Decision-Making through a Real-Effort Threshold Public Goods Game

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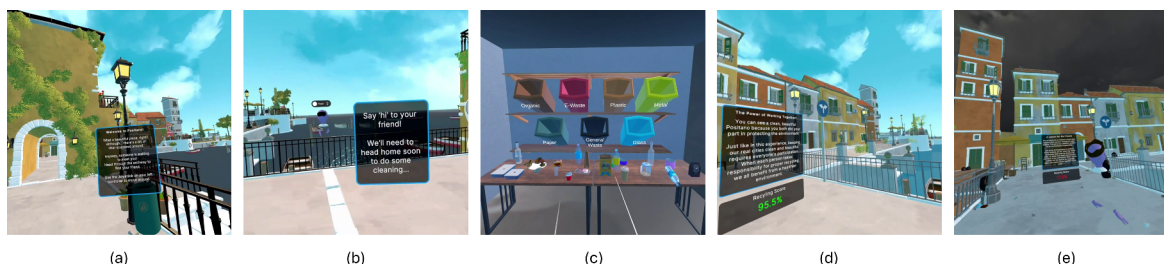


Figure 1: (a) Subjects are immersed in the nice Positano (Italy). (b) Each one observes the peer's representation; they can talk to each other, thus increasing the sense of being together in the shared VE. (c) Subjects proceed to a two-player threshold public goods game, framed as a real-effort recycling task. They have the common objective of reaching the residual waste target and preserving Positano's beauty. (d) If the common threshold is reached, Positano will be sunny, nice and clean; (e) otherwise Positano will be dirty and with a threatening storm.

ABSTRACT

We describe our contribution to the IEEE 3DUI Contest for IEEE VR 2025. Specifically, we present a collaborative Virtual Environment (VE) to be used as an innovative method to conduct experimental research in behavioral economics studies. Indeed, in such a field, analyzing collaborative behaviors is crucial, and collaborative VEs might be exploited to better control the variables of interest while preserving a high sense of realism and presence, to spread collaboration across wider geographical areas, and to maintain anonymity while preserving the sense of being with peers inside environments. The developed system aims to be used as a general framework to perform a wide range of studies in the field, here explicitly focusing on a real-effort recycling task, where subjects decide whether to prioritize individual gains or contribute to a collective effort to meet a common recycling objective.

Index Terms: Collaborative Virtual Environments, Threshold Public Goods Game, Residual Municipal Waste Target

1 INTRODUCTION

This contribution to the IEEE 3DUI context investigates the impact of Virtual Reality (VR) on economic decision-making, focusing on how a digital representation of subjects influences cooperative behavior. Specifically, this demo aims to represent the everyday challenge of citizens who have to make an effort (e.g., in terms of time) to correct recycle items, without having a direct advantage of doing it, but having in mind the "common good", i.e., the societal and environmental benefits of reaching the waste targets (see 2030

Agenda for Sustainable Development - Goal 12¹). Subjects begin the experiment by sharing a virtual space with their peers. This initial phase allows the players to observe their peer's representation, evoking the sense of the peer being present in the shared realistic environment. Then, subjects proceed to a two-player threshold public goods game, framed as a real-effort recycling task, where they decide whether to prioritize individual gains or contribute to a collective effort to meet a common recycling objective.

Here, we focus on a crucial advantage of VR when applied to behavioral economics studies: VR represents a valuable tool for experiments where context and realism can be added to the classical in-lab experimental tasks. The realism of the VR experiments is evident in three different ways, which can result in greater emotional investment in the task compared to abstract experimental models, where subjects often feel a psychological distance. First, the subjects in VR experiments have the opportunity to be immersed in decision-making situations that closely resemble those of the real world. Second, subjects can tangibly experience the consequences of their actions and the ones of their peers. Third, immersion in VR allows for a representation of themselves and other subjects taking part in the experiment. Regarding this third point, we emphasize that VR enables the exploration of scenarios in which the appearance of the subject and their peers—other subjects involved in the same task—is visually represented. This approach addresses a key limitation of traditional experimental procedures: anonymity. Experimental economics traditionally requires anonymity to ensure unbiased results. Through digital representations, VR allows subjects to remain anonymous while introducing a "humanization" form, thus enabling a unique balance between preserving anonymity and providing a visual representation of the other subjects.

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¹<https://sdgs.un.org/goals/goal12>

2 BACKGROUND

The use of VR to create simulators to teach positive behavior for environmental protection and waste reduction is quite diffuse in several recent works [6], because of the demonstrated effectiveness of VR for training and teaching. Moreover, collaborative VR is exploited in various fields, and many works focus on training [5]. In the context of experimental economics, we have two main methods: laboratory and field experiments, both of which have distinct advantages and disadvantages [1]. VR offers a novel approach to conducting experiments, enabling the creation of experimental settings that blend the advantages of both laboratory and field environments. These settings strike a balance between replicating real-life situations and maintaining experimental control and replicability [4, 3]. VR-based experiments allow researchers to design scenarios involving effort-based and interactive tasks [3, 2] in ways that mimic everyday actions, unlike traditional laboratory settings, which are often more abstract. Therefore, VEs offer context to experimental methods, raising the external validity of theoretical models and making findings more applicable to real-world situations [4, 2]. VR provides additional measures such as movement tracking, proxemics, and interpersonal distance behavior to complement traditional output measures to interpret economic decisions and behavioral patterns [2]. Human behavior in High-Immersive VEs media remains underexplored compared to Low-Immersive VEs in the economics literature [4]. The proposed collaborative VE contributes to the growing body of literature on VR in economics—an area still in its infancy—by specifically examining the effects of virtual representations and collaborative tasks of humans on economic behavior.

3 DESCRIPTION AND IMPLEMENTATION OF THE DEMO

The demo is a specific implementation of a more general collaborative VE designed to perform experiments in the context of real-effort threshold public goods game. The context, the task, and the controlled variables may vary according to the specific experimental design and user study.

Two subjects join a shared VE, which represents an enjoyable seaside town, namely Positano (Italy), because it is well-known across the world. In such a virtual town, subjects can explore the surroundings, meet the other players, and exchange words and greetings. This phase aims to enhance the sense of presence in the city and co-presence with others. After that, people leave the square in town and go into an indoor environment where they are introduced to the task: they are in a room with plenty of garbage and some recycling bins, and they have to clean up the room correctly, differentiating the items within a given time. Both subjects have to complete the same task, but at this stage, they cannot see each other. Like in the real world experience, several objects must be disassembled before recycling. An object can also be placed in a black bin for unsorted waste. The time limit and the number of items have been calibrated to make recycling challenging, again mimicking the real experience where time constraints and busy life may contrast with our willingness to be good citizens. On the other hand, the black bin for unsorted waste represents a less desirable but quick alternative. The two subjects have a common threshold to reach in terms of the percentage of correctly recycled items. Again, this is modeling the EU residual municipal waste target². If the subjects reach the target at the end of the collaborative game, they will see again a clean and nice Positano seaside square, otherwise they will see it full of garbage and with a threatening storm.

From the model's perspective, the utility of each user $i \in \{1, 2\}$

²<https://www.eea.europa.eu/publications/reaching-2030s-residual-municipal-waste>

can be represented as

$$\pi_i = \begin{cases} B + g \cdot y_i & \text{if } \frac{x_1 + x_2}{2} \geq T \\ g \cdot y_i & \text{otherwise.} \end{cases}$$

where $T \in [0, 1]$ is the common threshold of percentage of correctly recycled items for successful cooperation, and the vector (x_i, y_i) , with $x_i, y_i \in (0, 1)$ and $x_i + y_i = 1$, represents the percentage of correctly recycled items and the percentage of items placed in any other bin (including the black bin for unsorted waste, or an incorrect recycling bin), respectively. The value B represents the common utility being the result of a successful cooperation—the public goods utility—while g the unitary utility of being one step closer to the ultimate goal of cleaning up one's room.

It is worth noting that a user alone cannot reach the common threshold: subjects must collaborate to reach the target, as happens in our common life. Moreover, several factors have been implemented to mimic the current challenges in correctly recycling: objects are composed of different materials, and sometimes, it is not obvious to understand the correct bin, so the corresponding label must be localized.

The demo has been developed with Unity 2022.3.46f1 using the OpenXR Plugin 1.12.0 to make it consistent with the OpenXR standard. XR Interaction Toolkit 3.0.5 was used to develop interactions with objects in the VE. Netcode for GameObjects 1.11.0 was used to implement the necessary networking functionalities. To streamline the development, we used these packages together with Unity's VR Multiplayer Project Template that provides various Unity Cloud Multiplayer Services. We specifically utilized the Lobby 1.2.2 package from these services to implement quick join functionality, allowing players to rapidly create or join a game session and begin the collaborative experience.

4 FUTURE DEVELOPMENTS AND USAGE

The first straightforward outcome of this demo is the possibility of easily teaching unskilled people the main principles of public goods games, focusing on a topic of great importance at a societal level, i.e., environmental protection. Indeed, the two-player game directly shows the effect of collaborative behaviors and real-effort threshold in common yet essential actions, like correct recycling.

In a future step, subjects will begin the experiment by sharing a virtual space with their peers under one of three conditions: no avatar, an abstract avatar, or a human-like avatar. Thus, by examining how avatar realism affects social presence and collaboration, this research would provide valuable insights into the role of humanized digital peers in fostering prosocial behavior, offering a novel contribution to the literature on identification and social presence in economic experiments.

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